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# **COMMUNITY ISSUE REPORTING SYSTEM**

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### I. INTRODUCTION

- **Project Overview**

The Community Issue Reporting System is a digital platform designed to help people report barangay problems in their community area easily. It focuses on improving how communities handle common issues and making the reporting process of the problem more organized and accessible for the barangay to handle and solve the problem as fast as possible.

Users can register or log in and submit reports about community problems like illegal garbage disposal, road damage, flooding, etc. Each report includes details of the problem such as description, category, location, and an optional photo for the prof. The system helps barangay staff manage these reports by assigning priority levels and tracking their status. It also includes features like a history board, automatic sorting, notifications, and a search function. Admins can view all reports through a dashboard and see basic analytics. This project also supports SDG 11 "Sustainable Cities and Communities" which aims to make communities safer and more organized.

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The project helps improve communication between residents and local authorities. It makes reporting easier and helps communities respond to problems faster and more effectively.

- **UN SDG Target**

The Community Issue Report System supports the goals of the United Nations (UN) SDG 11 “Sustainable Cities and Communities” by helping improve how local problems in the community are reported and managed by the organization like barangay.

Through the system residents can quickly report issues like loitering, flooding, and road damage, while barangay staff can organize and respond to these reports using priority levels and status tracking. This directly supports 11.3 by encouraging community participation and 11.6 by helping reduce environmental problems.

By making reporting faster, more organized and easier to track, the project helps create a cleaner, safer, and more responsive community.

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- **Connection of the System to the UN SDG**

The Community Issue Reporting System is directly connected to the goals of the United Nations SDG 11: Sustainable Cities and Communities because it helps solve everyday problems in the community in a more organized way.

The system allows residents to report issues like garbage, flooding, and damaged infrastructure, making it easier for local authorities to take action. Features like priority levels, status tracking, and report history help ensure that problems are handled properly and not ignored.

By improving communication between residents and barangay staff, the system supports a cleaner environment, safer public spaces, and better community management. This shows how technology can be used to achieve the goals of SDG 11 in a practical and effective way.

## II. REQUIREMENT ANALYSIS

### List of Functional and Non-Functional Requirements

This section presents the functional and non-functional requirements of the Community Issue Reporting System. These requirements define the system's

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expected behavior, features, and quality attributes necessary to ensure effective community issue reporting and management aligned with Sustainable Development Goal (SDG) 11.

### A. Functional Requirements

Functional requirements specify the essential system functions that allow users and administrators to interact with the Community Issue Reporting System.

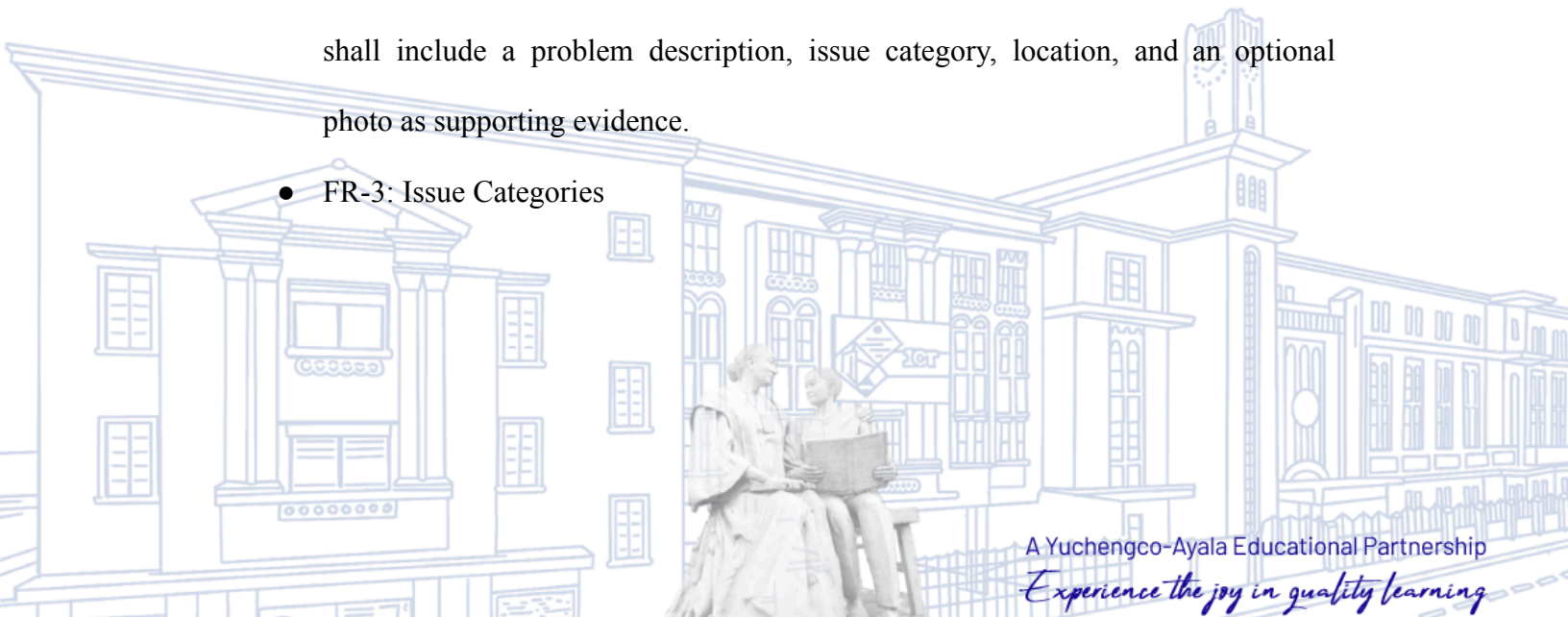
- FR-1: User Registration and Authentication

The system shall allow residents to register by providing valid user information. Registered users shall be able to log in and log out securely. Only authenticated users shall be permitted to submit and manage issue reports.

- FR-2: Issue Report Submission

The system shall enable users to submit community issue reports. Each report shall include a problem description, issue category, location, and an optional photo as supporting evidence.

- FR-3: Issue Categories



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The system shall support predefined issue categories, including but not limited to garbage disposal, road damage, flooding, street light malfunction, and noise complaints.

- FR-4: User Report Management

The system shall allow users to view their submitted reports. Users shall be able to edit or delete reports provided the report status remains pending.

- FR-5: Administrative Report Management

The system shall allow authorized barangay staff to access all submitted reports. Administrators shall be able to assign priority levels (Low, Medium, High, or Urgent) and update report statuses (Pending, In Progress, or Resolved).

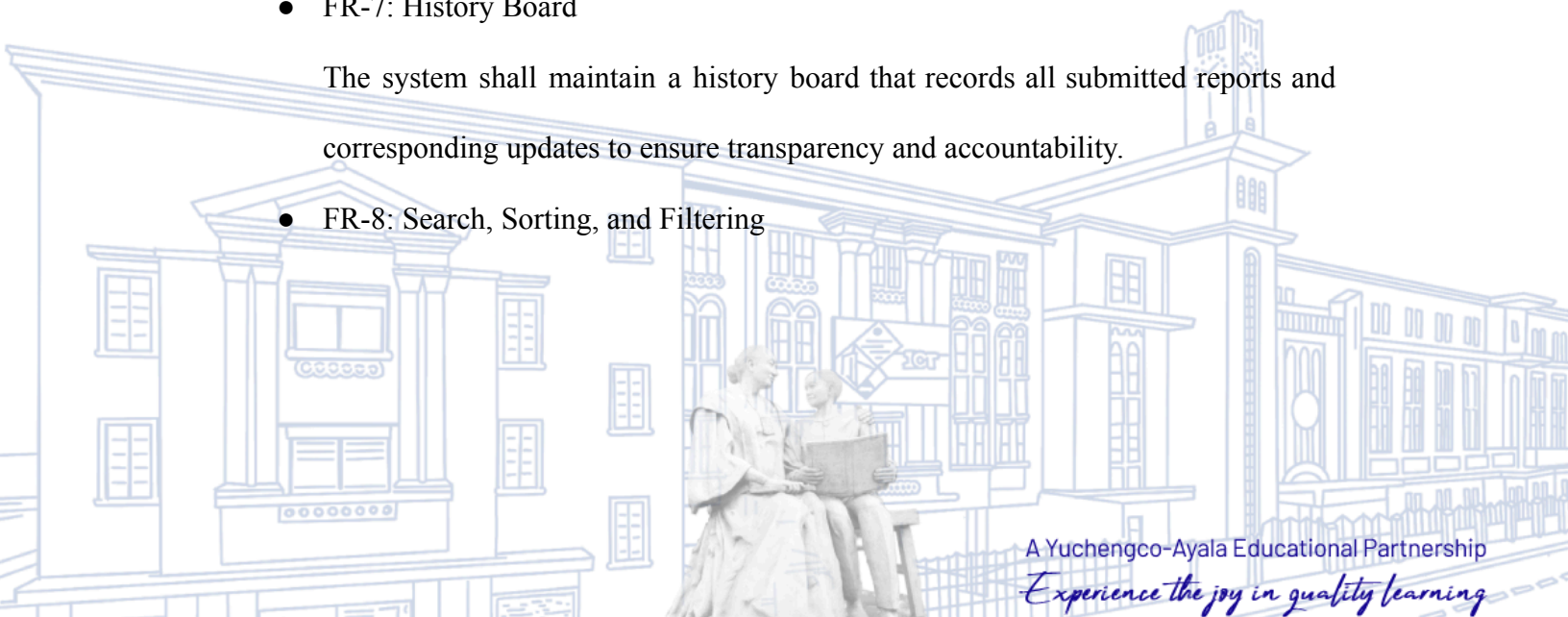
- FR-6: Assignment to Authorities

The system shall allow administrators to assign reported issues to appropriate barangay personnel or authorities for resolution.

- FR-7: History Board

The system shall maintain a history board that records all submitted reports and corresponding updates to ensure transparency and accountability.

- FR-8: Search, Sorting, and Filtering



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The system shall provide automatic sorting of reports based on category, priority level, status, and date. A search function shall be available to facilitate quick retrieval of reports.

- FR-9: Notification System

The system shall notify users when there are changes in the status or priority of their submitted reports.

- FR-10: Dashboard and Analytics

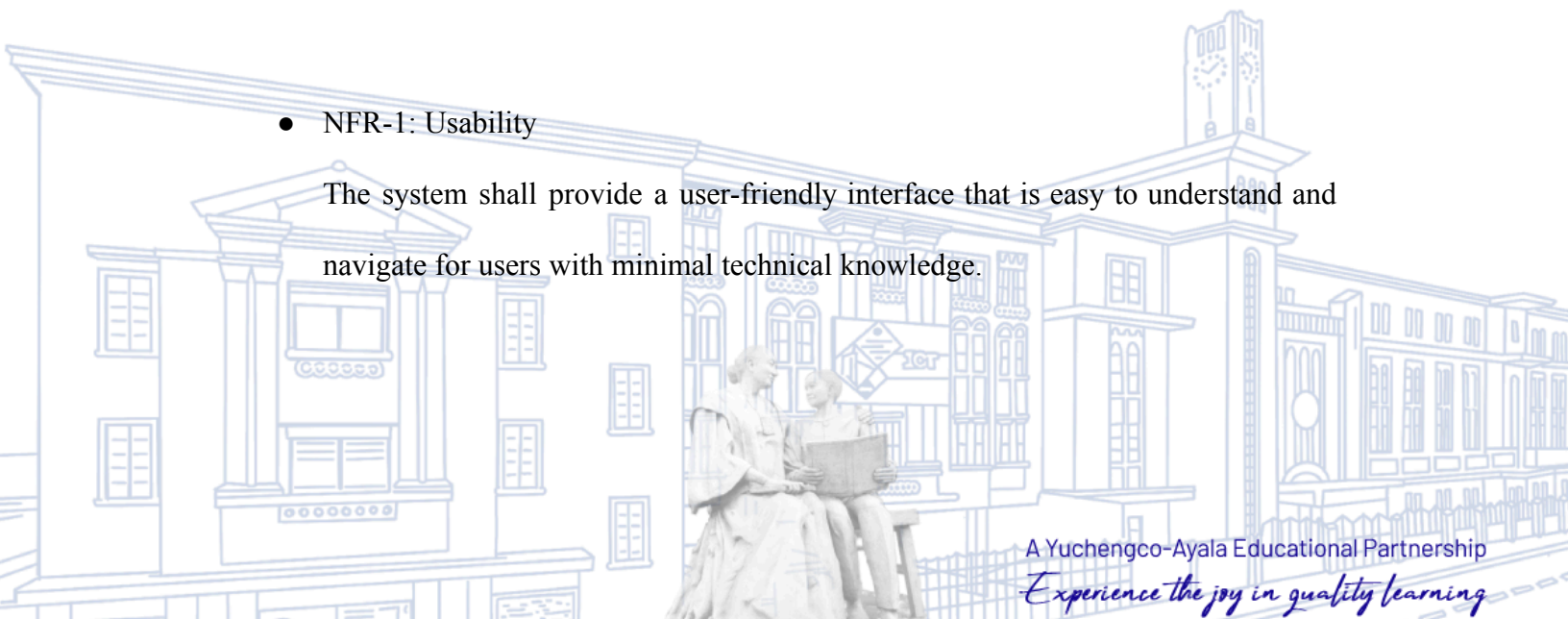
The system shall provide an administrative dashboard displaying the total number of reports, as well as counts of pending, in-progress, and resolved cases. Basic analytics showing the most frequently reported issues shall also be generated.

### B. Non-Functional Requirements

Non-functional requirements describe the quality attributes and constraints under which the system must operate.

- NFR-1: Usability

The system shall provide a user-friendly interface that is easy to understand and navigate for users with minimal technical knowledge.



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- NFR-2: Performance

The system shall process user requests and load system pages within an acceptable response time under normal operating conditions.

- NFR-3: Reliability

The system shall ensure data accuracy and integrity. All submitted reports and user information shall be stored reliably without data loss.

- NFR-4: Security

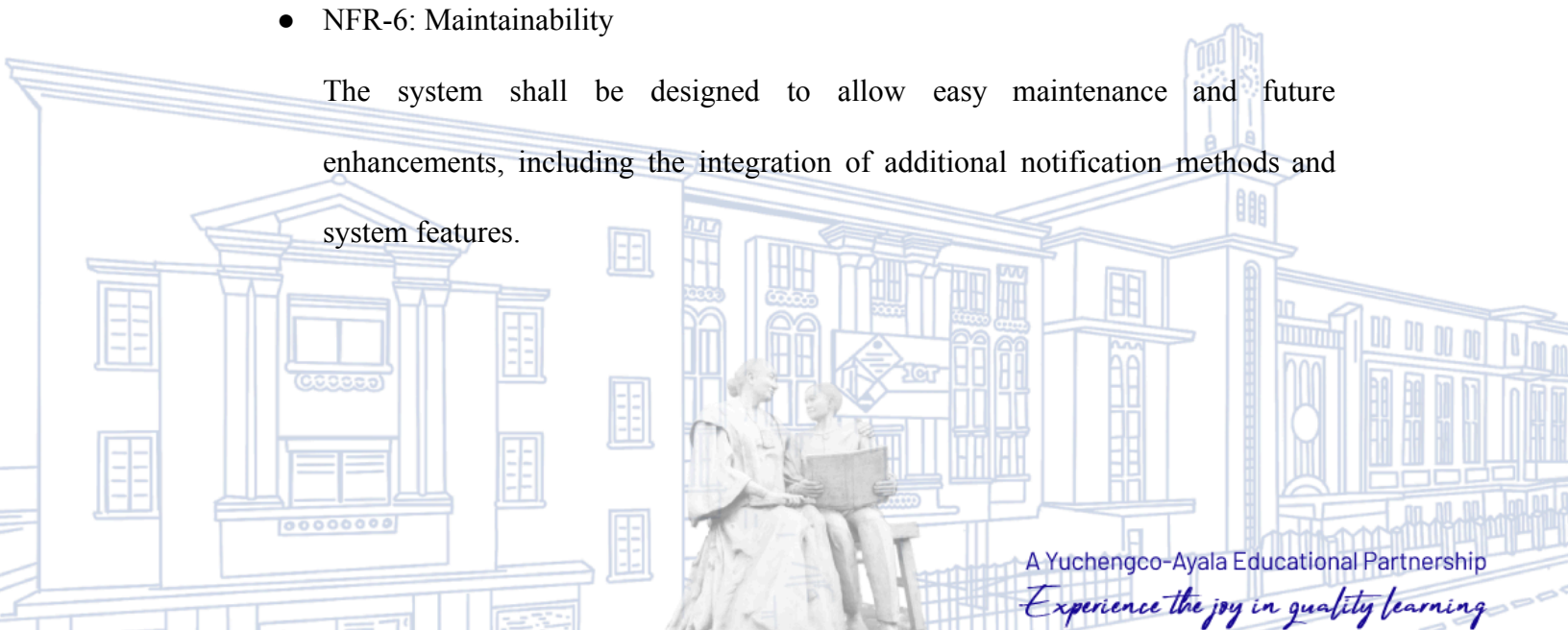
The system shall implement secure authentication mechanisms to protect user accounts. Access to administrative features shall be restricted to authorized personnel only.

- NFR-5: Scalability

The system shall be capable of accommodating an increasing number of users and reports as the community grows.

- NFR-6: Maintainability

The system shall be designed to allow easy maintenance and future enhancements, including the integration of additional notification methods and system features.



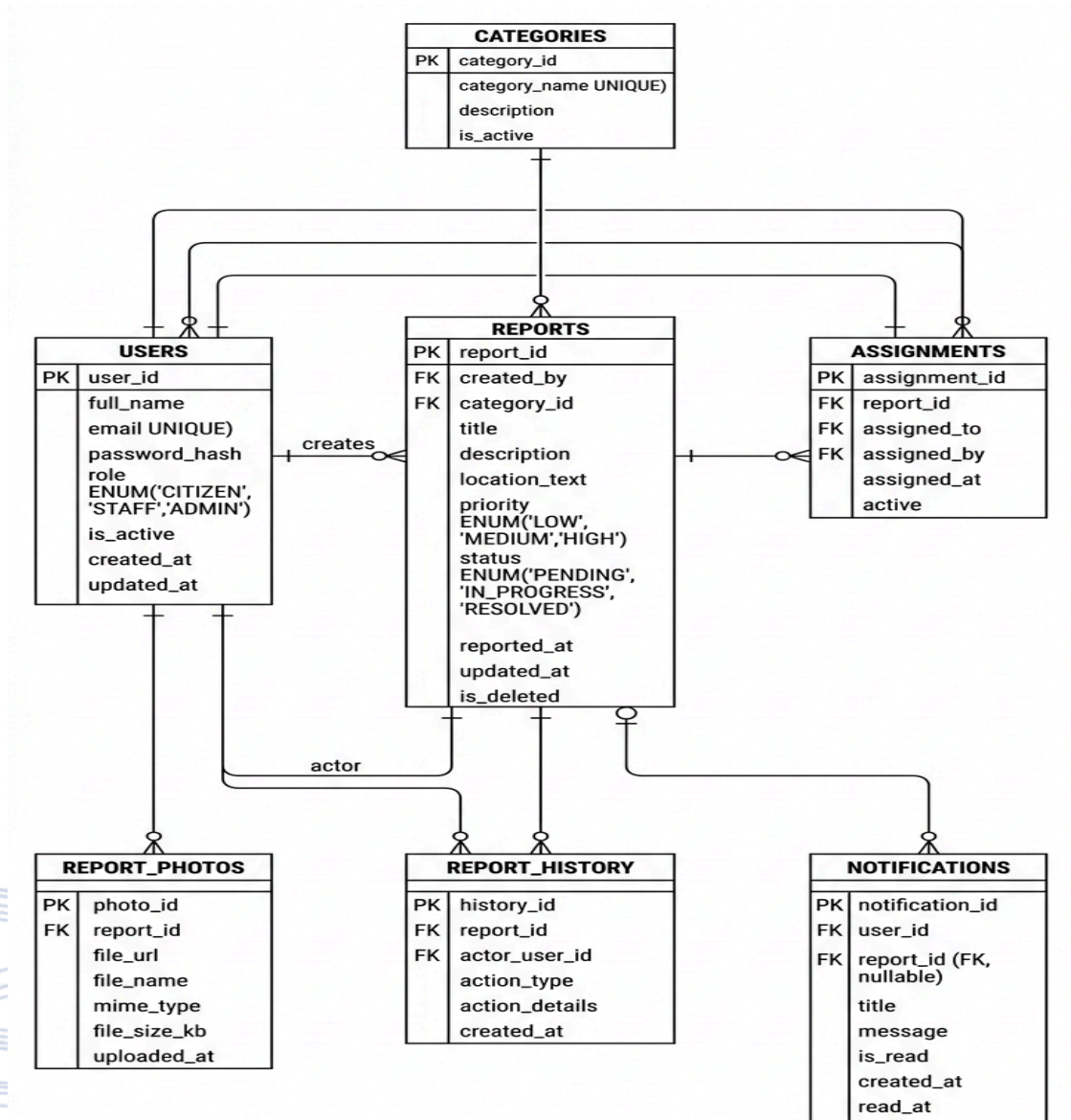
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### III. DATABASE DESIGN

- Entity Relation Diagram (ERD)



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## • Data Dictionary (Table Names, Fields, Types, and Constraints)

| Table Name     | Field Name       | Data Type     | Constraints                  | Description                                                                 |
|----------------|------------------|---------------|------------------------------|-----------------------------------------------------------------------------|
| USERS          | user_id          | INT           | PK, AUTO_INCREMENT           | Unique identifier for user                                                  |
| USERS          | username         | VARCHAR(50)   | UNIQUE, NOT NULL             | Login username                                                              |
| USERS          | email            | VARCHAR(100)  | UNIQUE, NOT NULL             | User email address                                                          |
| USERS          | password_hash    | VARCHAR(255)  | NOT NULL                     | Encrypted password                                                          |
| USERS          | role             | ENUM          | NOT NULL, DEFAULT 'resident' | User role: resident or admin                                                |
| USERS          | phone            | VARCHAR(20)   | NULL                         | Contact phone number                                                        |
| USERS          | created_at       | TIMESTAMP     | DEFAULT CURRENT_TIMESTAMP    | Account creation timestamp                                                  |
| USERS          | updated_at       | TIMESTAMP     | ON UPDATE CURRENT_TIMESTAMP  | Last update timestamp                                                       |
| REPORTS        | report_id        | INT           | PK, AUTO_INCREMENT           | Unique identifier for report                                                |
| REPORTS        | user_id          | INT           | FK → USERS(user_id)          | Reference to user who submitted                                             |
| REPORTS        | category         | ENUM          | NOT NULL                     | Issue category: garbage, road_damage, flooding, street_lights, noise, riots |
| REPORTS        | title            | VARCHAR(200)  | NOT NULL                     | Report title                                                                |
| REPORTS        | description      | TEXT          | NULL                         | Detailed description                                                        |
| REPORTS        | location_address | VARCHAR(255)  | NULL                         | Location address text                                                       |
| REPORTS        | latitude         | DECIMAL(10,8) | NULL                         | GPS latitude                                                                |
| REPORTS        | longitude        | DECIMAL(11,8) | NULL                         | GPS longitude                                                               |
| REPORTS        | photo_url        | VARCHAR(500)  | NULL                         | Photo file URL                                                              |
| REPORTS        | priority_level   | ENUM          | DEFAULT 'low'                | Priority: low/medium/high/urgent                                            |
| REPORTS        | status           | ENUM          | DEFAULT 'pending'            | Status: pending/in_progress/resolved                                        |
| REPORTS        | assigned_to      | INT           | FK → USERS(user_id)          | Admin assigned to resolve                                                   |
| REPORTS        | created_at       | TIMESTAMP     | DEFAULT CURRENT_TIMESTAMP    | Report creation timestamp                                                   |
| REPORTS        | updated_at       | TIMESTAMP     | ON UPDATE CURRENT_TIMESTAMP  | Last update timestamp                                                       |
| STATUS_HISTORY | history_id       | INT           | PK, AUTO_INCREMENT           | Unique identifier for history entry                                         |
| STATUS_HISTORY | report_id        | INT           | FK → REPORTS(report_id)      | Reference to report                                                         |
| STATUS_HISTORY | old_status       | VARCHAR(50)   | NOT NULL                     | Previous status value                                                       |
| STATUS_HISTORY | new_status       | VARCHAR(50)   | NOT NULL                     | New status value                                                            |
| STATUS_HISTORY | changed_by       | INT           | FK → USERS(user_id)          | User who made the change                                                    |
| STATUS_HISTORY | changed_at       | TIMESTAMP     | DEFAULT CURRENT_TIMESTAMP    | When change occurred                                                        |
| STATUS_HISTORY | notes            | TEXT          | NULL                         | Additional notes                                                            |
| NOTIFICATIONS  | notification_id  | INT           | PK, AUTO_INCREMENT           | Unique identifier for notification                                          |
| NOTIFICATIONS  | user_id          | INT           | FK → USERS(user_id)          | Recipient user                                                              |
| NOTIFICATIONS  | report_id        | INT           | FK → REPORTS(report_id)      | Related report                                                              |
| NOTIFICATIONS  | type             | ENUM          | NOT NULL, DEFAULT 'push'     | Notification type: email/sms/push                                           |
| NOTIFICATIONS  | message          | TEXT          | NOT NULL                     | Notification message content                                                |
| NOTIFICATIONS  | is_read          | BOOLEAN       | DEFAULT FALSE                | Read status flag                                                            |
| NOTIFICATIONS  | sent_at          | TIMESTAMP     | DEFAULT CURRENT_TIMESTAMP    | When notification sent                                                      |
| HISTORY_BOARD  | board_id         | INT           | PK, AUTO_INCREMENT           | Unique identifier for board entry                                           |
| HISTORY_BOARD  | report_id        | INT           | FK → REPORTS(report_id)      | Related report                                                              |
| HISTORY_BOARD  | action_type      | VARCHAR(50)   | NOT NULL                     | Type of action performed                                                    |
| HISTORY_BOARD  | action_by        | INT           | FK → USERS(user_id)          | User who performed action                                                   |
| HISTORY_BOARD  | action_at        | TIMESTAMP     | DEFAULT CURRENT_TIMESTAMP    | When action occurred                                                        |
| HISTORY_BOARD  | details          | TEXT          | NULL                         | Action details                                                              |
| ANALYTICS_LOG  | log_id           | INT           | PK, AUTO_INCREMENT           | Unique identifier for analytics entry                                       |
| ANALYTICS_LOG  | report_id        | INT           | FK → REPORTS(report_id)      | Related report                                                              |
| ANALYTICS_LOG  | category         | VARCHAR(50)   | NOT NULL                     | Issue category for analytics                                                |
| ANALYTICS_LOG  | priority         | VARCHAR(20)   | NOT NULL                     | Priority level for analytics                                                |
| ANALYTICS_LOG  | status           | VARCHAR(20)   | NOT NULL                     | Status for analytics                                                        |
| ANALYTICS_LOG  | logged_at        | TIMESTAMP     | DEFAULT CURRENT_TIMESTAMP    | When logged                                                                 |

Color Key:  Primary Key (PK)  Foreign Key (FK)  USERS Table  REPORTS Table  NOTIFICATIONS  ANALYTICS\_LOG

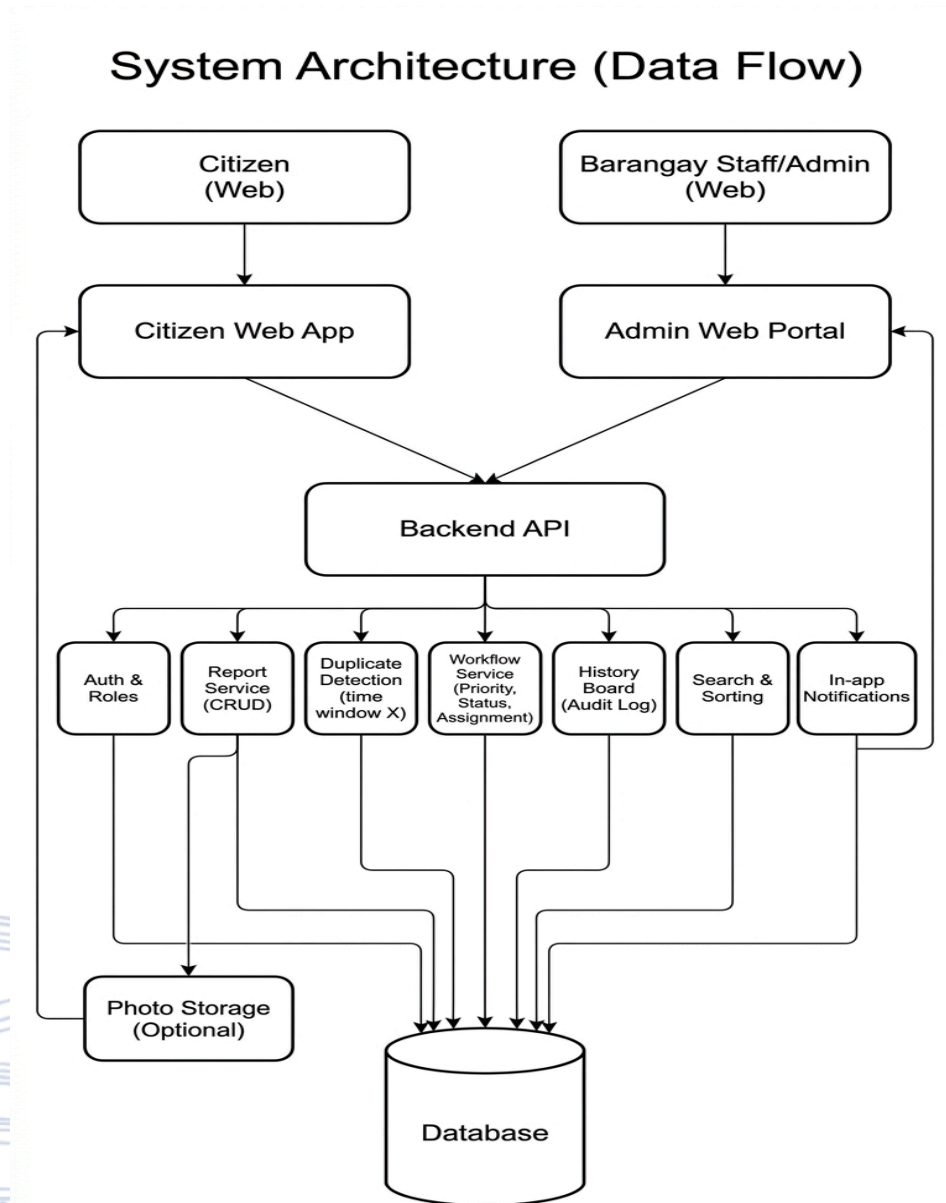
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### IV. SYSTEM ARCHITECTURE

- Data Flow



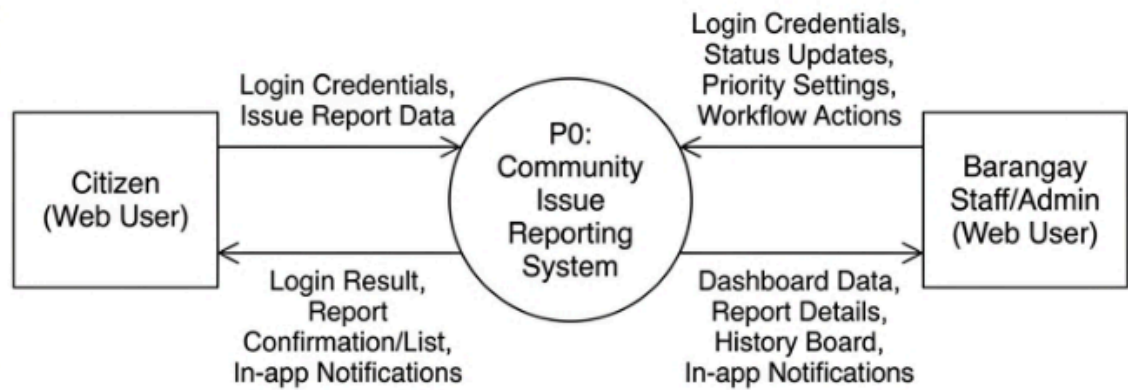




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- DFD Level 0 and 1

### DFD Level 0



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```

    usecaseDiagram
        actor Citizen as Citizen (Web User)
        actor Staff as Barangay Staff/Admin (Web User)
        actor D1 as D1 Users
        usecase U1 as 1.0 User Mgmt (Auth/Roles)
        usecase U2 as 2.0 Issue Report Mgmt (CRUD)
        usecase U3 as 3.0 Duplicate Detection
        usecase U4 as 4.0 Workflow Mgmt
        usecase U5 as 5.0 History Board Logging
        usecase U6 as 6.0 Search & Sorting
        usecase U8 as 8.0 Dashboard & Analytics
        dataStore D1 as D1 Users
        dataStore D2 as D2 Reports
        dataStore D3 as In-app Notifications
        dataStore D4 as D4 History Logs
        dataStore D5 as D5 Notifications
        dataStore D6 as D6 Config/Reference (Categories, Status, Priority, Duplicate Window X)

        Citizen --> U1
        Citizen --> U2
        Citizen --> U6
        Citizen --> U8
        Staff --> U1
        Staff --> U2
        Staff --> U6
        Staff --> U8
        U1 <--> D1
        U1 --> D6
        U2 --> U3
        U2 --> U4
        U2 --> U5
        U3 <--> D2
        U4 <--> D2
        U5 <--> D2
        D2 <--> D3
        D3 --> D4
        D4 --> D5
        D5 --> U6
        U6 --> Citizen
        U6 --> U8
        U8 --> Citizen
        U8 --> Staff
  
```



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### V. INDIVIDUAL CONTRIBUTION

Wireframe & User Interface – Vince Alex M. Alcala & Junior C. Delos Santos

System Flow & Code – Renz Christian M. Malco & Rehan T. Alamada

Documentation – Sophia Bea U. Rivera & Lance Andrei A. Gulila

### IV. AI-ASSISTED TOOLS USED IN THE STUDY

- ChatGPT (OpenAI) – Assisted in system design guidance, CRUD logic explanation, debugging support, and documentation clarity.
- Gemini (Google) – Used for information gathering and concept verification.
- Grammarly (Grammarly, Inc.) – Used for grammar checking and sentence improvement.



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